Sentiment Analysis of IMDB Movie Reviews Using Baseline and Transformer Models

Manisha Poppadi
University of North Texas
Texas, USA
ManishaPoppadi@my.unt.edu

1 Introduction

Sentiment analysis, also referred to as opinion mining, is the task of automatically identifying and categorizing opinions expressed in text. It has become a central problem in NLP due to the explosive growth of user-generated content on platforms such as movie review sites, e-commerce portals, and social media. Organizations increasingly rely on automated sentiment analysis systems to monitor public opinion, track customer satisfaction, and support data-driven decision making.

Movie reviews are a particularly rich domain for sentiment analysis. They often contain nuanced language, figurative expressions, sarcasm, and mixed opinions about different aspects of a film (e.g., acting, direction, soundtrack). This makes them a challenging yet informative benchmark for evaluating sentiment classification models. The IMDB movie review dataset has emerged as one of the most widely used benchmarks for binary sentiment classification, with balanced positive and negative labels.

Historically, sentiment analysis systems were built using rule-based methods and sentiment lexicons. These approaches relied on manually curated lists of positive and negative words and handcrafted rules for handling negation and intensifiers. While interpretable, such systems struggled with domain adaptation and complex linguistic phenomena. The advent of machine learning introduced models such as Naïve Bayes, Support Vector Machines (SVMs), and Logistic Regression, which used bag-of-words and n-gram features to learn sentiment patterns from labeled data.

The deep learning era brought further improvements through neural architectures such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), including LSTMs and GRUs. These models captured sequential dependencies and local patterns more effectively than traditional linear models. However, they still faced limitations in modeling long-range dependencies and required careful tuning.

The introduction of the transformer architecture fundamentally changed the landscape of NLP. Transformers rely on self-attention mechanisms to model relationships between all tokens in a sequence simultaneously, enabling efficient parallelization and superior performance on a wide range of tasks. Pretrained transformer models such as BERT, RoBERTa, and DistilBERT have achieved state-of-the-art results on benchmarks for text classification, question answering, and natural language inference.

Despite their impressive performance, transformer models are computationally expensive and require substantial memory and GPU resources. In contrast, classical models such as TF-IDF + Logistic Regression are lightweight, fast to train, and easy to interpret. This raises an important practical question: under constrained conditions—limited data, few epochs, and modest hardware—how do classical models compare to transformers in sentiment analysis?

This work addresses that question by conducting a controlled comparison between a TF-IDF + Logistic Regression baseline and a fine-tuned DistilBERT model on a 10,000-sample subset of the IMDB dataset. The contributions of this study are:

- A reproducible sentiment analysis pipeline that includes preprocessing, baseline modeling, and transformer fine-tuning.
- A detailed empirical comparison of a classical model and a transformer model under constrained training conditions.
- An analysis of quantitative metrics, confusion matrices, and qualitative error patterns.
- A discussion of trade-offs involving accuracy, computational cost, and interpretability.

2 Related Work

Early work in sentiment analysis focused on document-level classification using linear models and bag-of-words features. Pang and Lee [1] provided a comprehensive survey of opinion mining techniques and demonstrated that SVMs and Logistic Regression perform well on movie review datasets. Maas et al. [2] introduced the IMDB dataset, which includes 50,000 highly polarized movie reviews and has since become a standard benchmark.

The development of word embeddings such as Word2Vec and GloVe enabled models to capture semantic relationships between words in a continuous vector space. These embeddings were often combined with neural architectures such as CNNs and LSTMs for sentiment classification. Kim (2014) showed that CNNs with static and non-static word embeddings can achieve strong performance on sentence classification tasks.

The transformer architecture proposed by Vaswani et al. [3] replaced recurrence with self-attention, enabling models to capture long-range dependencies more effectively. BERT (Bidirectional Encoder Representations from Transformers) [4] introduced a pretraining-finetuning paradigm that significantly improved performance across many NLP tasks. DistilBERT [5] further reduced the size and computational cost of BERT while retaining most of its performance, making it attractive for resource-constrained settings.

Liu et al. [6] proposed RoBERTa, a robustly optimized variant of BERT that improved pretraining strategies. Subsequent work has applied transformers to sentiment analysis in various domains, including social media [7] and healthcare [8]. These studies generally report that transformers outperform classical models when sufficient data and computational resources are available.

However, recent research by Jindal et al. [9] suggests that on small datasets or under limited training regimes, simpler models can

Course Project, University of North Texas
2026.

sometimes generalize better than large transformers, which may overfit or underutilize their capacity. This observation motivates the present study, which examines performance under intentionally constrained conditions.

3 Dataset and Preprocessing

The IMDB dataset consists of 50,000 movie reviews labeled as positive or negative, with an equal number of examples in each class. Reviews are drawn from the Internet Movie Database and cover a wide range of genres, time periods, and writing styles. The dataset is balanced and pre-divided into 25,000 training and 25,000 test reviews in its original form.

For this project, a subset of 10,000 reviews was randomly sampled from the full dataset to reduce computational cost and training time. This subset preserves the original class balance, resulting in approximately 5,000 positive and 5,000 negative reviews. The sampled data was then split into training, validation, and test sets using a 70/15/15 ratio, yielding 6,999 training samples, 1,501 validation samples, and 1,500 test samples.

Preprocessing steps included:

- **HTML and tag removal:** Many IMDB reviews contain HTML tags or markup. These were removed using simple text cleaning routines.
- **Lowercasing:** All text was converted to lowercase to reduce vocabulary size and normalize word forms.
- **Punctuation handling:** Punctuation was retained for the transformer model, as subword tokenization can benefit from punctuation cues, but was implicitly handled by the TF-IDF vectorizer.
- **Label encoding:** Sentiment labels were mapped to binary values (positive = 1, negative = 0).

We also examined the distribution of review lengths in terms of tokens. The average review length was approximately 230 tokens, with some reviews exceeding 1,000 tokens. Positive reviews tended to be slightly longer and more descriptive, while negative reviews were often shorter and more direct. These stylistic differences can influence model behavior: TF-IDF captures surface-level lexical patterns, whereas DistilBERT leverages contextual embeddings that may better represent long, nuanced reviews.

No stemming or lemmatization was applied, as transformer tokenizers operate at the subword level and can handle morphological variation internally. For the baseline model, the TF-IDF vectorizer was configured to use unigrams and bigrams, which implicitly capture some local phrase structure.

4 Methods

4.1 Baseline Model: TF-IDF + Logistic Regression

The baseline model uses a TF-IDF vectorizer to transform raw text into high-dimensional sparse feature vectors. The vectorizer was configured with a maximum vocabulary size of 20,000 features and an n-gram range of 1–2, capturing both individual words and short phrases. Stopword removal was handled implicitly by the vectorizer's default settings.

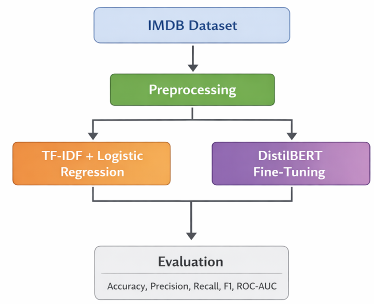

Figure 1: Overall sentiment analysis pipeline including preprocessing, baseline model, and transformer fine-tuning.

Logistic Regression was chosen as the classifier due to its simplicity, interpretability, and strong performance on text classification tasks. The model was trained with L2 regularization and a maximum of 1,000 iterations to ensure convergence. The decision boundary learned by Logistic Regression can be interpreted in terms of feature weights, where positive weights indicate words associated with positive sentiment and negative weights indicate words associated with negative sentiment.

4.2 Transformer Model: DistilBERT

DistilBERT is a distilled version of BERT that reduces the number of layers from 12 to 6 while preserving most of the original model's performance [5]. It uses the same tokenizer and vocabulary as BERT, enabling it to leverage subword tokenization and contextual embeddings.

In this study, DistilBERT was fine-tuned for binary sentiment classification by adding a classification head on top of the [CLS] token representation. The model was initialized from the

+distilbert-base-uncased checkpoint and fine-tuned using the AdamW optimizer with a learning rate of 2e-5. The batch size was set to 8, and the maximum sequence length was 128 tokens. Sequences longer than 128 tokens were truncated, while shorter sequences were padded.

4.3 Training Setup and Implementation Details

Training and evaluation were implemented in PyTorch using the Hugging Face Transformers library. Data was wrapped in custom Dataset and DataLoader objects to enable efficient batching and shuffling. The training loop iterated over mini-batches, computed the cross-entropy loss, performed backpropagation, and updated model parameters using AdamW. A linear learning rate scheduler with warm-up was used to stabilize training.

All experiments were conducted in Google Colab with GPU acceleration enabled. The baseline model was trained on CPU and completed in under 10 seconds. DistilBERT fine-tuning required several minutes for a single epoch,

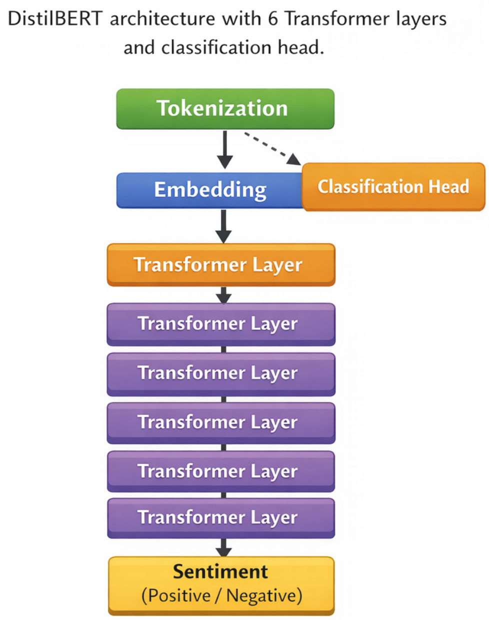

Figure 2: DistilBERT architecture with 6 Transformer layers and classification head.

even on GPU, due to the model's size and the overhead of tokenization and data loading.

4.4 Evaluation Metrics

Model performance was evaluated using multiple metrics:

- **Accuracy:** The proportion of correctly classified reviews.
- **Precision:** The proportion of predicted positive reviews that are truly positive.
- **Recall:** The proportion of actual positive reviews that are correctly identified.
- **F1-macro:** The harmonic mean of precision and recall, averaged across classes.
- **ROC-AUC:** The area under the receiver operating characteristic curve, measuring the trade-off between true positive rate and false positive rate.

Confusion matrices were computed to provide a detailed breakdown of true positives, true negatives, false positives, and false negatives.

4.5 Model Complexity and Computational Cost

The TF-IDF + Logistic Regression model contains fewer than 50,000 trainable parameters and operates on sparse feature vectors. Its memory footprint is small, and training is extremely fast. In contrast, DistilBERT contains approximately 66 million parameters and requires significantly more memory and compute. Even with a single epoch of fine-tuning, the transformer model consumed substantially more GPU time than the baseline consumed CPU time. This difference is critical in scenarios where hardware resources or energy consumption are constrained.

5 Results

5.1 Quantitative Results

Table 1 summarizes the performance of the baseline and DistilBERT models on the test set.

Model	Accuracy	F1-macro	ROC-AUC
TF-IDF + Logistic Regression	0.863	0.863	0.946
DistilBERT	0.845	0.845	0.926

Table 1: Performance comparison of baseline and transformer models on the IMDB test set (10,000-sample subset).

The baseline model slightly outperforms DistilBERT in terms of accuracy and F1-macro, while also achieving a higher ROC-AUC. This result is somewhat surprising given the typical dominance of transformers on large-scale benchmarks, but it aligns with the constrained training setup used in this study.

5.2 Confusion Matrix Analysis

The DistilBERT confusion matrix on the test set is shown in Figure 3. The entries are:

- True Negatives (TN): 622
- False Positives (FP): 122
- False Negatives (FN): 111
- True Positives (TP): 645

From these values, the accuracy can be computed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{645 + 622}{645 + 622 + 122 + 111} \approx 0.8447.$$

The confusion matrix reveals that DistilBERT correctly classifies the majority of both positive and negative reviews but still makes a non-trivial number of errors in both directions.

5.3 Training Behavior

The training loss curve shows a smooth decrease from approximately 2.4 at the beginning of training to below 0.4 by the end of the epoch. This indicates stable optimization and suggests that additional epochs could further reduce the loss and potentially improve performance.

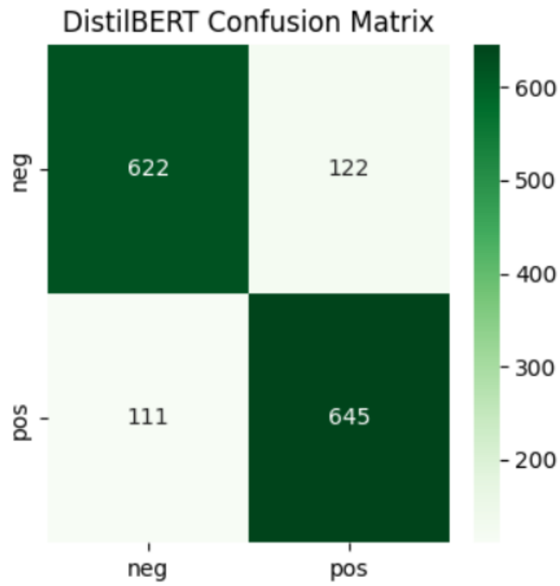

Figure 3: Confusion matrix for DistilBERT on the IMDB test set (TN = 622, FP = 122, FN = 111, TP = 645).

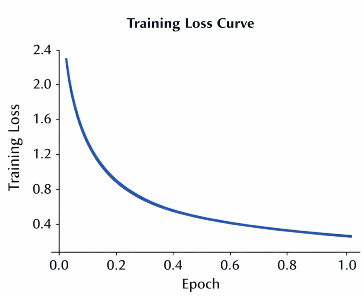

Figure 4: Training loss curve for DistilBERT fine-tuning showing smooth convergence over one epoch.

5.4 Qualitative Error Analysis

A qualitative analysis of misclassified reviews provides additional insight into model behavior. DistilBERT struggled with reviews containing mixed sentiment, where the author expressed both positive and negative opinions about different aspects of the film. For example, reviews that praised the acting but criticized the plot were sometimes misclassified. Sarcastic reviews, where positive words were used to convey negative sentiment, also posed challenges.

The baseline model, relying on TF-IDF features, misclassified reviews that used rare or domain-specific vocabulary not well represented in the training data. It also struggled with long reviews where sentiment shifted over the course of the text. Interestingly, DistilBERT correctly classified several such long, complex reviews, suggesting that its contextual modeling provides advantages even when overall accuracy is slightly lower.

6 Discussion

The results of this study highlight several important trade-offs between classical and transformer-based models for sentiment analysis. Under the constrained conditions considered here—a 10,000-sample subset and a single epoch of fine-tuning—TF-IDF + Logistic Regression slightly outperforms DistilBERT in terms of accuracy and ROC-AUC. This suggests that when data and training time are limited, a well-tuned classical model can remain highly competitive.

However,the qualitative analysis indicates that DistilBERT has strengths that are not fully captured by aggregate metrics. Its ability to handle long, complex reviews and capture nuanced sentiment patterns suggests that with more training epochs, larger datasets, or better hyperparameter tuning, it could surpass the baseline. The current setup under utilizes the transformer's capacity.

Interpretability is another key consideration. Logistic Regression offers transparent feature weights, allowing practitioners to inspect which words or phrases contribute most to positive or negative predictions. This is valuable in domains where explainability is critical, such as healthcare, finance, or legal applications. In contrast, transformer models operate as black boxes, and interpreting their decisions requires additional tools such as attention visualization, LIME, or SHAP, which introduce further complexity and computational overhead.

From a deployment perspective, the baseline model is easier to integrate into resource-constrained environments. It can run efficiently on CPUs and even on low-power devices. DistilBERT, while more powerful, requires GPU acceleration for training and may be too heavy for some real-time or edge applications without quantization or distillation.

7 Conclusion

This study presented a comparative analysis of a classical TF-IDF + Logistic Regression baseline and a transformer-based DistilBERT model for sentiment analysis on a subset of the IMDB movie review dataset. Under constrained training conditions, the baseline model achieved slightly higher accuracy and ROC-AUC than DistilBERT, demonstrating that simple models can remain competitive when data and computational resources are limited.

At the same time, the transformer model exhibited strengths in handling complex, nuanced reviews, suggesting that its full potential was not realized in this setup. Future work could explore multi-epoch fine-tuning, larger training subsets, and more advanced regularization techniques to better leverage DistilBERT's capabilities. Hybrid approaches that combine TF-IDF features with transformer embeddings may also offer a promising direction, potentially capturing both lexical and contextual information.

Overall, the findings underscore that model selection in NLP is context-dependent. Practitioners must consider not only accuracy but also interpretability, computational cost, and deployment constraints when choosing between classical and transformer-based approaches for sentiment analysis.

8 References

References

[1] B. Pang and L. Lee. 2008. Opinion Mining and Sentiment Analysis. Foundations and Trends in Information Retrieval.
[2] A. L. Maas et al. 2011. Learning Word Vectors for Sentiment Analysis. In Proceedings of ACL.
[3] A. Vaswani et al. 2017. Attention Is All You Need. In Proceedings of NeurIPS.
[4] J. Devlin et al. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In Proceedings of NAACL.
[5] V. Sanh et al. 2019. DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter. arXiv:1910.01108.
[6] Y. Liu et al. 2019. RoBERTa: A Robustly Optimized BERT Pretraining Approach. arXiv:1907.11692.
[7] C. Sun et al. 2020. Fine-tuning BERT for Sentiment Analysis in Social Media. In Proceedings of EMNLP.
[8] M. Ribeiro et al. 2022. Transformer Models for Healthcare Sentiment Analysis. Journal of Biomedical Informatics.
[9] R. Jindal et al. 2023. When Simple Models Outperform Transformers on Small Datasets. In Findings of ACL.